4(b). QUEUE USING LINKED LIST PROGRAM

```
#include <stdio.h>
#include <stdlib.h>

// Structure to create a node with data and next pointer
struct node {
	int data;
	struct node *next;
};

struct node *front = NULL;
struct node *rear = NULL;

// Enqueue() operation on a queue
void enqueue(int value) {
	struct node *ptr;
	ptr = (struct node *)malloc(sizeof(struct node));
	ptr->data = value;
	ptr->next = NULL;
	if ((front == NULL) && (rear == NULL)) {
		front = rear = ptr;
	} else {
		rear->next = ptr;
		rear = ptr;
	}
	printf("Node is Inserted:%d \n\n",value);
}

// Dequeue() operation on a queue
int dequeue() {
	if (front == NULL) {
		printf("\nUnderflow\n");
		return -1;
	} else {
		struct node *temp = front;
		int temp_data = front->data;
		front = front->next;
		free(temp);
		return temp_data;
	}
}

// Display all elements of queue
void display() {
	struct node *temp;
	if ((front == NULL) && (rear == NULL)) {
```

```
                printf("\nQueue is Empty\n");
        } else {
                printf("The queue is \n");
                temp = front;
                while (temp) {
                        printf("%d--->", temp->data);
                        temp = temp->next;
                }
                printf("NULL\n\n");
        }
}


int main()
{
        enqueue(10);
        enqueue(20);
        enqueue(30);
        display();
        dequeue();
        display();
}
```